Chicken and Egg:

Should you use the VIX to time the SPX?

Or use the SPX to time the VIX?

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3/24/2024

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Abstract

"Chicken and Egg: Should you use the VIX to time the SPX? Or use the SPX to time the VIX?" presents an exploration into the reciprocal predictive relationships between the S&P 500 Index (SPX) and the CBOE Volatility Index (VIX), with a particular focus on deriving actionable insights for market timing strategies. This comprehensive study challenges the prevailing market wisdom by positing that SPX action offers a more reliable basis for forecasting VIX movement than VIX action does in forecasting SPX movement.

The introduction sets the stage by questioning the traditional reliance on the VIX as a market timing tool, suggesting instead that SPX movements may offer valuable insights into future VIX trends. Numerous works are cited sampling publicly available research. Most past research has utilized VIX as an indicator for SPX. The study probes deeper to understand the extent to which one index can predict the movements of the other.

The paper then explains the history and construction of the VIX. It shows how the VIX has moved during all major market selloffs since 1990, which is as far back as our VIX data is available.

The next section is a discussion of implied and historical volatility (HV). It begins with a dialogue of the “Rule of 16”, and what different VIX readings actually represent. This section lays the groundwork for better understanding the interaction between SPX and VIX, and why the movements of SPX influence option traders and their willingness to pay more for portfolio protection. As it turns out, options traders are more likely to be reactionary than anticipatory.

At this point, the paper examines VIX versus SPX readings as short-term market indicators. Two types of studies are done here. One uses short-term RSI readings of 2, 3, and 4 days. The other looks at short-term high and low readings, from 5 to 25 days in length. Readings are calculated on SPX and also on VIX. Data is then examined which shows the SPX readings tend to be more useful in anticipating SPX movement than VIX readings are. The

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data is sliced several ways, and the answer consistently suggests SPX readings offer more robust edges for trading.

There is then a discussion of VIX-based securities and why traders and investment managers may want to consider them in a portfolio. One of the primary reasons demonstrated is that they have shown an incredible downside persistency. While counter moves can be sharp during market crises, VIX-based securities (futures and exchange-traded products) have typically gone substantially shorter periods between making new all-time lows than SPX has in making new all-time highs. The Great Financial Crisis of 2007-2008 along with the 2022 bear market are discussed in detail. Analysis implies that a well-constructed short-VIX strategy should have an edge over a long-SPX strategy when looking to avoid lengthy periods of drawdown.

With this in mind, studies are shown that are constructed in a similar manner to those that examined SPX movements. The paper again utilizes short-term RSI readings and short-term high-low readings of both SPX and VIX. This time it examines their value as indicators for VIX futures movements. Again it is found that SPX readings tend to act as a superior indicator. Data here is also broken down to examine behavioral differences when SPX is in a long-term uptrend (above its 200ma) versus a long-term downtrend (below its 200ma).

While the data tables show strong evidence that SPX readings could be used effectively to find favorable opportunities for shorting VIX futures, the final section takes it a bit further. To show the consistency that the edge has exhibited over time, a simple model is built using some of the concepts and indicators previously examined. The model makes it clear that the research is capable of serving as a core component in winning strategies that could be developed for investors.

Final analysis declares that SPX indicators, particularly overbought/oversold measures, possess a markedly higher predictive value for VIX movements than previously recognized.

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The paper challenges conventional wisdom and presents a compelling case for the predictive superiority of SPX over VIX in the context of trading both SPX and VIX-based securities. This sets the stage for further research while providing traders and investment managers tools to realize untapped potential in leveraging SPX action for more informed and effective trading decisions in the volatility space.

Introduction

Do bigger chickens produce bigger eggs? Do bigger eggs lead to bigger chickens? We might not know whether the chicken or the egg came first, but we can probably learn the answer to the above. When looking at the market, we know that the S&P 500 (SPX) came before the CBOE Volatility Index (VIX). But which does a better job of predicting the other? And more importantly, where do the biggest trading edges lie?

There has been ample research published over the years that examined CBOE Volatility Index (VIX) action, and how VIX readings might be used to predict S&P 500 movement. In their 2004 book, "How Markets Really Work", Larry Connors and Conor Sen showed how short-term overbought and oversold VIX readings were a useful indicator for short-term SPX movement. Connors has since published additional research along these lines.

In 2018, Fassas and Hourvouliades published "VIX Futures As A Market Timing Indicator" which examined VIX futures term structure for anticipating forward S&P 500 returns.

In his 2020 NAAIM Founders Award paper, "Actively Using Passive Sectors to Generate Alpha Using the VIX" Michael Gayed used the VIX in timing allocation adjustments between sectors.

Studies have also been published that examine short volatility strategies, such as "VIX Index Strategies, Shorting volatility as a portfolio enhancing strategy", by Dondoni, Montagna, and Maggi (2018). There they identified the long-term tendency of VIX futures to decline and used the VIX futures term structure to aid in timing short volatility positions.

A primary use of VIX among researchers and market participants has been as an indicator for timing the SPX and other markets. Years of trading experience and research has taught me that more value may be found in doing the converse. Therefore, I will examine

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approaches that use SPX action as an indicator for trading VIX-based securities, primarily VIX futures. Before getting to that, I will discuss the VIX itself, what its readings represent, and historical tendencies of the VIX. I'll then examine whether VIX readings are any more predictive of SPX movement than simply using SPX readings. I'll examine VIX futures (VX) movement, the use of SPX indicators as a VX trading filter, and finally, a simple model that incorporates some of the research.

VIX Overview

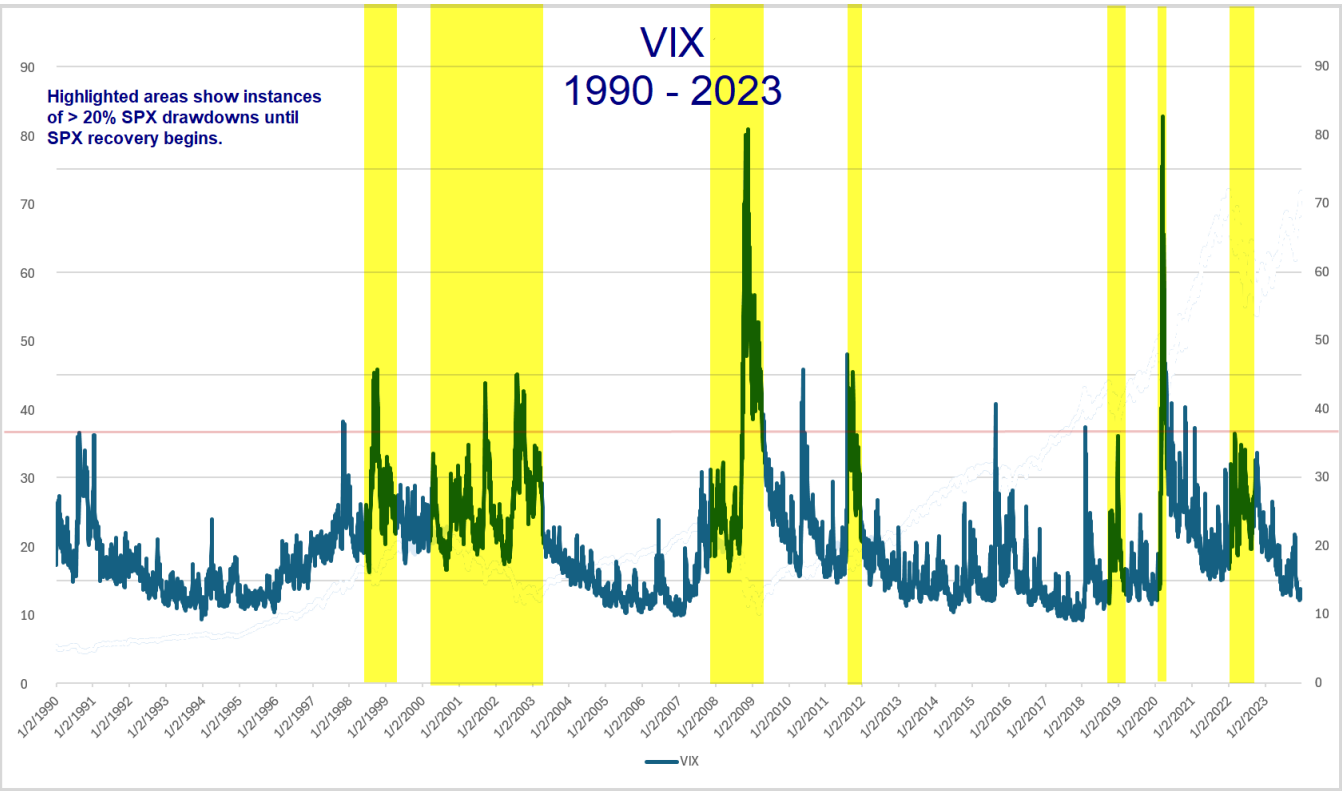
The VIX was created by the Chicago Board of Options Exchange (CBOE). It measures 30-day S&P 500 volatility expectations based on SPX option prices. To get the 30-day average, it considers option prices in the 23 to 37-day range. The VIX will generally deliver "high" readings in volatile and uncertain conditions. "Low" readings are often found during quiet market environments. From 1990 – 2023 the range of closing VIX prices was between 9.14 and 82.69.

The VIX is not tradeable, but there are tradeable instruments based on the VIX. VIX futures were the first instrument to be based on the VIX index. They began trading in 2004, though volume was quite muted in the early years. In 2006 VIX options were introduced. Then in 2009, we saw the first VIX-based ETFs arrive. Interest in using VIX-based instruments has grown substantially over the years. For some they are used as a hedging instrument, and for others they are traded as a separate strategy.

The reason VIX products became a popular hedge is that the VIX will often trade inversely to the SPX. The basic reason for this is that falling markets will generate fear. The desire for protection increases and traders are generally more willing to pay a larger premium for options to protect their portfolios. When SPX is on the rise, investors feel good. There is less demand for portfolio protection. People are not willing to pay as much for options in a rising market as in a falling market. So the reduced options premium is reflected in the lower VIX reading.

Below is a long-term chart of daily VIX closing prices. Though SPX is not shown on the chart, yellow highlighted sections are times where the SPX experienced a 20% drawdown or more.

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VIX spikes occurred during all of the tumultuous S&P 500 drawdowns, including the 1998 Long-Term Capital blowup, the 2000-2003 tech bear, the 2008 Great Financial Crisis, the 2020 COVID Crash, and others. But as you can see, the VIX tended to revert to more normal levels quite quickly, even when the S&P 500 took an extended amount of time to recover its losses. Quicker recovery time is a key concept. And it not only applies to the VIX, but as I will show later, it also applies to tradeable VIX-based products.

I will also share another observation here. The 2022 high level is shown by the red line. Along with the December 2018 bear, this is about as low a reading as we have seen during a bear market. That surprised some people, but it shouldn't have. There was nothing particularly scary about the 2022 bear market. It was primarily caused by a rise in interest rates. Rising interest rates are not nearly as scary as a worldwide financial crisis or a 100-year pandemic. Further, realized volatility never spiked in 2022 like we saw in other periods. So with actual price movement not extreme and the "big scare" being rising interest rates, why would people expect the VIX to reach the kind of levels that were seen in some other bear markets? They shouldn't.

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Implied & Historical Volatility

I think part of the problem with people misunderstanding where the VIX "should" be is that they do not understand implied volatility, and they especially don't understand what the VIX readings represent. Implied volatility is an estimate of daily variations in price. The VIX measures options that are priced on the S&P 500. It is the implied volatility of those SPX options that average 30 days to maturity. But it is not a "daily" number. It is "annualized". This leads us to the "Rule of 16".

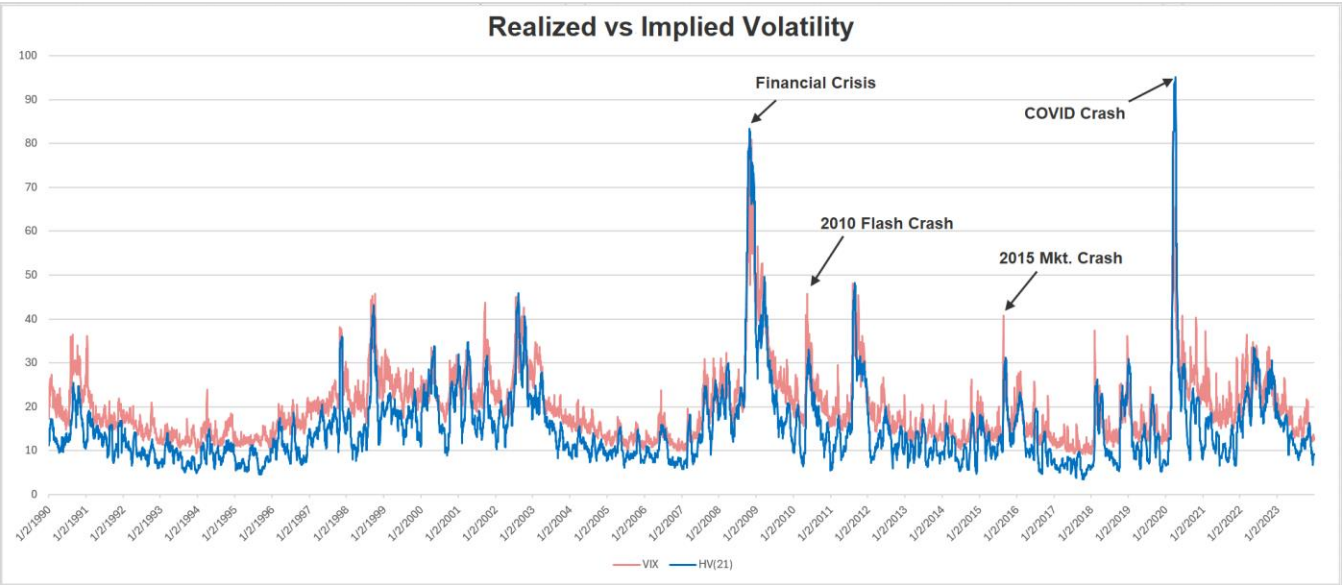
To convert an annual implied volatility number into a daily number you need to take the square root of the number of days. There are approximately 252 trading days in a year. The square root of 252 is 15.875 (about 16). So a VIX reading of 16 represents daily implied volatility of about 1% over the next 30 days. (More specifically, it suggests that 68.2% of days will see price changes of less than 1%, and 31.8% of the time the SPX could change by more than 1%.) A VIX of 32 implies 2% moves are likely, 48 would anticipate numerous 3% moves, and for the VIX to be at 80, implied volatility on SPX options would be pricing in 5% daily moves.

If you have observed the S&P 500 for very long, you will realize that price changes of 5%, 4%, 3%, or even 2% in one day are quite rare. Going through an extended period of time where daily moves are averaging even 2% is highly unusual. So it is understandable that a VIX of even 30 or 40 is not common.

When price action in the S&P 500 becomes more volatile, the VIX therefore naturally spikes. And when S&P 500 volatility begins to quiet down, it is unlikely that the VIX will remain elevated.

When considering what volatility expectations might be over the next month, the easiest way to estimate this would be to look at what volatility has been over the last month. One month forward is rounded to 30 days for the VIX. One month backward can be rounded to 21 (trading) days for historical volatility (HV). The chart below shows VIX readings in red and 21-day HV of SPX in blue, going back to the inception of the VIX in 1990.

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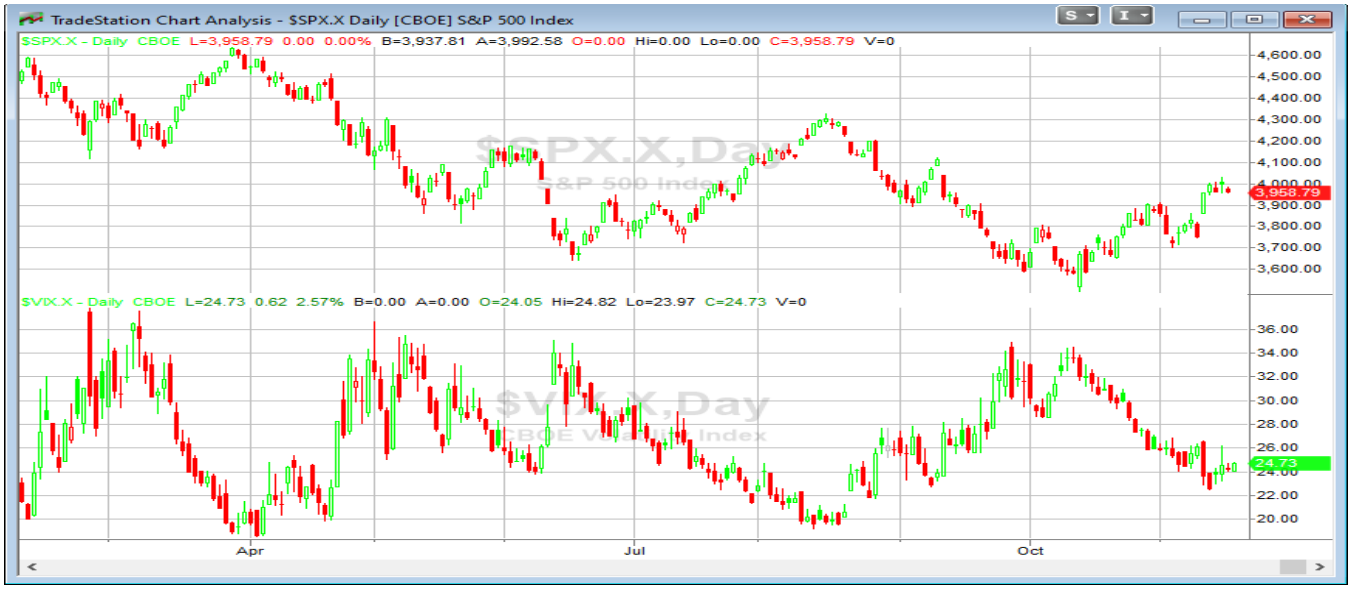
A few notes:

- Perhaps the most apparent observation I could make about the chart is that the lines tend to move together. They rise and fall at about the same time, never getting too far off track.
- The red line (VIX) is typically above the blue (HV). So estimations of implied volatility are generally a little higher than what has been seen from a volatility standpoint in the recent past. This makes sense. The next 30 days' action is unknown, and options sellers will typically be able to demand a premium for taking on unknown risks.
- There are a couple of instances marked on the chart that HV reached similar levels to the 2022 highs, but VIX spiked quite a bit higher. They were in 2010 and 2015. Important to note about both these instances is that they were quick market shocks. HV transitioned from a low level to up over 30 very quickly both times. In other words, there was a brief but powerful market shock that VIX quickly adjusted to, and this caused it to elevate more substantially above HV.
- The 2022 high levels in HV were achieved through a more gradual rise. Options traders had more time to adjust to increased volatility, and so VIX remained more in line with HV as compared to the 2010 and 2015 instances.

So VIX is a measure of implied volatility for S&P 500 options. It is not related to actual, historical volatility, but the two are highly correlated.

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I mentioned the VIX often trades inversely to the SPX. Let's look at a chart so you can see what I mean by this.



This chart is from 2022. You can see here how moves up in SPX generally coincide with moves down in the VIX, and vice versa. Because VIX direction is often inverted versus SPX, the tendency over the years has been to use VIX as an indicator for SPX. Let's take a quantitative look to evaluate whether VIX is likely a useful indicator for SPX movement. I will look at it from both a long-term and a short-term perspective.

VIX vs SPX Readings as a Long-Term Market Filter

I do not recall reading any research in the past that used VIX as an effective long-term indicator. Of course SPX long-term trend measures and long-term moving averages can be used to potentially sidestep portions of bear markets, reduce drawdown, and in many cases increase long-term returns.

So I took 3 simple S&P 500 long-term trend filters to see how utilizing them might compare to a Buy & Hold approach with the S&P 500. The 3 trend filters I chose were:

1) *Whether the SPX closed higher than it closed 1 year ago (252 trading days)*
2) *Whether the SPX closed above its 200-day moving average.*
3) *Whether the 50-day moving average closed above the 200-day moving average (also known as the Golden Cross formation).*

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The test assumed that when the trend filter suggested an uptrend that SPX was entered. When SPX was not in an uptrend, then the portfolio would go to cash, and earn interest at the overnight Fed Funds rate.

Lastly, I took the same indicators and applied them to the VIX. Since the VIX typically mirrors the SPX, I reversed them. So the portfolio would go long if:

1) *The VIX closed below where it did 1 year ago.*
2) *The VIX closed below its 200-day moving average.*
3) *The 50-day moving average of the VIX closed below the 200-day moving average.*

Results using these six filters, along with "Buy & Hold" returns can be found in the table below.

SPX Performance Based on Long-Term Filters Using SPX or VIX.
$10,000 starting account. 1992 - 2023.

Long-Term Trend Id	# Trades	Net Profit	CAGR	Exposure %	Max. Sys % Drawdown	CAR/MDD
Buy & Hold	1	$104,359.73	7.91	100	-56.78	0.14
SPX Up Year	62	$106,382.64	7.97	78.73	-25.04	0.32
SPX > 200-day MA	114	$65,731.04	6.53	74.99	-22.1	0.3
SPX Golden Cross	15	$121,113.70	8.37	75.44	-33.92	0.25
VIX Down Year	292	$28,767.73	4.32	51.66	-38.65	0.11
VIX Below 200ma	320	$36,145.84	4.89	60.8	-36.94	0.13
VIX 50ma < 200ma	35	$41,683.38	5.26	60.48	-47.8	0.11

We see here that using SPX trend filters was generally successful. Two of the three SPX filters outperformed Buy & Hold total returns. They all reduced drawdowns. And the Compound Annual Return / Max Drawdown (CAR/MDD) stats were improved using all 3 filters.

But the long-term VIX filters did not seem to help anything. They all showed lower net profits, and the CAR/MDD was worse than Buy & Hold in every case. It appears long-term measures of the VIX are not helpful as a filter for SPX.

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VIX vs SPX Readings as a Short-Term Market Indicator

Most publicly available VIX-related research has focused on using VIX measures to anticipate SPX action. However, my research has not found VIX action to be any more predictive of SPX movement than simply using SPX action.

I'll use two different measures of overbought/oversold to demonstrate. The first indicator to consider is Wilder's RSI. Short-term measures of RSI are used here for a couple of reasons: 1) I have found that shorter-term RSIs tend to indicate short-term action better than longer-term RSIs. 2) Short-term RSIs tend to offer more evenly distributed delineation. For instance, a 14-period RSI will see most readings fall between 30 and 70. But a 2-period RSI will have plenty of instances below 30 and above 70. This makes for neater delineation.

The first series of tests utilize RSI(2). All of these tests are run from 2007 – 2023. This was done to include the Great Financial Crisis, and also to measure a period where both VIX futures and options were available. 2007 is the first full year where that was the case.

This first results table shows next-day returns in SPX when its RSI(2) closed in 5 different quintiles. It also incorporates a long-term filter (SPX 200-day moving average) for the bottom 10 rows.

Next Day SPX Performance Based on RSI(2) of SPX.
2007 - 2023.

Long-Term Status	Short-term Setup	rsiLength	# Trades	% of Winners	Avg % Profit/Loss	Profit Factor
0 Long-Term Filter	RSI <= 20	2	822	57.66	0.18	1.41
0 Long-Term Filter	RSI > 20 and <= 40	2	635	55.43	0.06	1.14
0 Long-Term Filter	RSI > 40 and <= 60	2	603	52.4	-0.04	0.93
0 Long-Term Filter	RSI > 60 and <= 80	2	851	55.23	0.03	1.08
0 Long-Term Filter	RSI > 80 and <= 100	2	1367	51.5	-0.03	0.91
Above 200-day MA	RSI <= 20	2	500	59.4	0.1	1.28
Above 200-day MA	RSI > 20 and <= 40	2	453	55.41	0.04	1.12
Above 200-day MA	RSI > 40 and <= 60	2	443	52.6	0.01	1.04
Above 200-day MA	RSI > 60 and <= 80	2	677	55.39	0.02	1.09
Above 200-day MA	RSI > 80 and <= 100	2	1129	52.35	0.01	1.03
Below 200-day MA	RSI <= 20	2	322	54.97	0.32	1.53
Below 200-day MA	RSI > 20 and <= 40	2	182	55.49	0.12	1.16
Below 200-day MA	RSI > 40 and <= 60	2	160	51.88	-0.17	0.82
Below 200-day MA	RSI > 60 and <= 80	2	174	54.6	0.05	1.07
Below 200-day MA	RSI > 80 and <= 100	2	238	47.48	-0.19	0.71

Results here show some interesting delineation, and are suggestive of an edge that would be worthwhile to consider. Low RSIs (especially below 20) provide the strongest numbers.

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Such oversold conditions led to net positive next-day returns. Meanwhile, overbought conditions (RSI > 80) showed negative returns. This was especially true during a long-term downtrends (bottom row). During long-term uptrends (above the 200ma), RSI>80 results were actually slightly positive. Also notable: 1) Some people may not be familiar with "Profit Factor". It is Gross Gains / Gross Losses. So a profit factor above 1 means that the strategy is profitable. A reading below 1 means the strategy has lost money over time. Profit factor can be a helpful measure of reward/risk when evaluating strategies. 2) Results when SPX is in a downtrend tend to be more extreme – both bullish and bearish. It may seem odd that average gains are much higher during "oversold in downtrends" than they are during "oversold in uptrends", but volatility tends to be higher during downtrends. When volatility is higher it makes moves more sizable. Hence the more extreme numbers during downtrends. This will be evident in results throughout the paper. 3) The theme of oversold providing an upside edge and overbought suggesting a neutral or bearish market condition is something you will continue to see throughout the research as well.

So now let's use the same RSI(2) indicator and apply it to the VIX to see if that might provide an edge for trading SPX.

Next Day SPX Performance Based on RSI(2) of VIX.
2007 - 2023.

Long-Term Status	Short-term Setup	rsiLength	# Trades	% of Winners	Avg % Profit/Loss	Profit Factor
0 Long-Term Filter	0 ST Filter	2	4278	54.79	0.04%	1.11
0 Long-Term Filter	RSI of VIX <= 20	2	1108	55.51	0.05%	1.14
0 Long-Term Filter	RSI of VIX > 20 and <= 40	2	914	54.81	0.01%	1.03
0 Long-Term Filter	RSI of VIX > 40 and <= 60	2	740	53.65	-0.02%	0.95
0 Long-Term Filter	RSI of VIX > 60 and <= 80	2	686	55.69	0.06%	1.14
0 Long-Term Filter	RSI of VIX > 80 and <= 100	2	830	54.1	0.13%	1.3
Above 200-day MA	0 ST Filter	2	3202	55.47	0.04%	1.13
Above 200-day MA	RSI of VIX <= 20	2	834	56.35	0.04%	1.17
Above 200-day MA	RSI of VIX > 20 and <= 40	2	708	55.93	0.04%	1.15
Above 200-day MA	RSI of VIX > 40 and <= 60	2	551	52.99	0.00%	0.99
Above 200-day MA	RSI of VIX > 60 and <= 80	2	499	58.32	0.09%	1.34
Above 200-day MA	RSI of VIX > 80 and <= 100	2	610	53.61	0.02%	1.07
Below 200-day MA	0 ST Filter	2	1076	52.79	0.06%	1.09
Below 200-day MA	RSI of VIX <= 20	2	274	52.92	0.07%	1.11
Below 200-day MA	RSI of VIX > 20 and <= 40	2	206	50.97	-0.09%	0.89
Below 200-day MA	RSI of VIX > 40 and <= 60	2	189	55.56	-0.08%	0.9
Below 200-day MA	RSI of VIX > 60 and <= 80	2	187	48.66	-0.03%	0.96
Below 200-day MA	RSI of VIX > 80 and <= 100	2	220	55.45	0.41%	1.66

Some of the same themes play out here, but not as impressively. An overbought VIX (RSI > 80, 0 Long-Term Filter, 6th row down) suggests the SPX may be primed to bounce. But an

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oversold VIX (RSI <= 20, 2nd row down) shows numbers that are very close to the numbers shown when there is no filter at all (top row). So while an overbought VIX could provide an edge, an oversold VIX does not appear to generate a quantifiable edge.

Let's next look at RSI(3) for SPX and VIX, in order to assess whether the themes play out here as well.

Next Day SPX Performance Based on RSI(3) of SPX.
2007 - 2023.

Long-Term Status	Short-term Setup	rsiLength	# Trades	% of Winners	Avg % Profit/Loss	Profit Factor
0 Long-Term Filter	RSI <= 20	3	457	59.52	0.29	1.6
0 Long-Term Filter	RSI > 20 and <= 40	3	782	54.22	0.04	1.09
0 Long-Term Filter	RSI > 60 and <= 80	3	1161	53.83	0.03	1.08
0 Long-Term Filter	RSI > 40 and <= 60	3	944	54.66	-0.02	0.96
0 Long-Term Filter	RSI > 80 and <= 100	3	934	51.28	-0.02	0.92
Above 200-day MA	RSI <= 20	3	248	62.5	0.2	1.58
Above 200-day MA	RSI > 60 and <= 80	3	917	54.31	0.02	1.08
Above 200-day MA	RSI > 40 and <= 60	3	683	54.76	0.02	1.06
Above 200-day MA	RSI > 20 and <= 40	3	531	54.8	0.01	1.04
Above 200-day MA	RSI > 80 and <= 100	3	823	52.13	0.01	1.03
Below 200-day MA	RSI <= 20	3	209	55.98	0.39	1.61
Below 200-day MA	RSI > 20 and <= 40	3	251	52.99	0.11	1.15
Below 200-day MA	RSI > 60 and <= 80	3	244	52.05	0.05	1.08
Below 200-day MA	RSI > 40 and <= 60	3	261	54.41	-0.13	0.84
Below 200-day MA	RSI > 80 and <= 100	3	111	45.05	-0.22	0.62

Once again, an RSI of SPX under 20 suggests a short-term bullish edge, and an RSI over 80 suggests a neutral (during uptrends) or bearish (during downtrends) edge.

Next let's apply RSI(3)of VIX as a delineator.

Next Day SPX Performance Based on RSI(3) of VIX.
2007 - 2023.

Long-Term Status	Short-term Setup	rsiLength	# Trades	% of Winners	Avg % Profit/Loss	Profit Factor
0 Long-Term Filter	RSI of VIX <= 20	3	569	54.31	0.02%	1.05
0 Long-Term Filter	RSI of VIX > 20 and <= 40	3	1292	56.81	0.07%	1.22
0 Long-Term Filter	RSI of VIX > 40 and <= 60	3	1121	51.92	-0.04%	0.9
0 Long-Term Filter	RSI of VIX > 60 and <= 80	3	836	56.34	0.09%	1.2
0 Long-Term Filter	RSI of VIX > 80 and <= 100	3	460	53.91	0.15%	1.34
Above 200-day MA	RSI of VIX <= 20	3	418	55.74	0.05%	1.2
Above 200-day MA	RSI of VIX > 20 and <= 40	3	1014	56.9	0.05%	1.2
Above 200-day MA	RSI of VIX > 40 and <= 60	3	841	52.44	-0.02%	0.95
Above 200-day MA	RSI of VIX > 60 and <= 80	3	590	57.8	0.09%	1.33
Above 200-day MA	RSI of VIX > 80 and <= 100	3	339	54.28	0.04%	1.1
Below 200-day MA	RSI of VIX <= 20	3	151	50.33	-0.07%	0.88
Below 200-day MA	RSI of VIX > 20 and <= 40	3	278	56.47	0.15%	1.25
Below 200-day MA	RSI of VIX > 40 and <= 60	3	280	50.36	-0.13%	0.85
Below 200-day MA	RSI of VIX > 60 and <= 80	3	246	52.85	0.07%	1.09
Below 200-day MA	RSI of VIX > 80 and <= 100	3	121	52.89	0.47%	1.74

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Here again, an overbought VIX suggests an upside edge for SPX, but an oversold VIX does not appear greatly predictive. An oversold VIX when SPX is in a downtrend does show slightly bearish results, but not nearly to the extent that an overbought RSI(3) of SPX does.

Finally…RSI(4). First, SPX:

Next Day SPX Performance Based on RSI(4) of SPX.
2007 - 2023.

Long-Term Status	Short-term Setup	rsiLength	# Trades	% of Winners	Avg % Profit/Loss	Profit Factor
0 Long-Term Filter	RSI <= 20	4	285	60	0.31	1.63
0 Long-Term Filter	RSI > 20 and <= 40	4	820	54.39	0.05	1.1
0 Long-Term Filter	RSI > 40 and <= 60	4	1145	54.24	0.02	1.05
0 Long-Term Filter	RSI > 60 and <= 80	4	1405	54.59	0.01	1.03
0 Long-Term Filter	RSI > 80 and <= 100	4	623	49.92	-0.02	0.91
Above 200-day MA	RSI <= 20	4	140	62.14	0.23	1.61
Above 200-day MA	RSI > 20 and <= 40	4	520	56.92	0.04	1.1
Above 200-day MA	RSI > 60 and <= 80	4	1142	55.25	0.02	1.09
Above 200-day MA	RSI > 40 and <= 60	4	821	53.59	0.02	1.06
Above 200-day MA	RSI > 80 and <= 100	4	579	50.6	0.01	1.03
Below 200-day MA	RSI <= 20	4	145	57.93	0.39	1.64
Below 200-day MA	RSI > 20 and <= 40	4	300	50	0.08	1.1
Below 200-day MA	RSI > 40 and <= 60	4	324	55.86	0.03	1.04
Below 200-day MA	RSI > 60 and <= 80	4	263	51.71	-0.04	0.93
Below 200-day MA	RSI > 80 and <= 100	4	44	40.91	-0.38	0.37

Again low RSI readings suggest an upside edge and high RSI readings suggest a next-day downside edge. Results here are a bit more extreme. Part of the reason for this is that it takes larger moves to get extreme readings with a 4-period RSI than a 2-period RSI. That is why you also see the lower number of instances where RSI was below 20 or above 80.

And VIX…

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Next Day SPX Performance Based on RSI(4) of VIX.
2007 - 2023.

Long-Term Status	Short-term Setup	rsiLength	# Trades	% of Winners	Avg % Profit/Loss	Profit Factor
0 Long-Term Filter	RSI of VIX <= 20	4	253	50.99	-0.07%	0.8
0 Long-Term Filter	RSI of VIX > 20 and <= 40	4	1443	55.99	0.06%	1.19
0 Long-Term Filter	RSI of VIX > 40 and <= 60	4	1435	53.94	0.01%	1.02
0 Long-Term Filter	RSI of VIX > 60 and <= 80	4	862	55.22	0.06%	1.13
0 Long-Term Filter	RSI of VIX > 80 and <= 100	4	285	55.09	0.23%	1.51
Above 200-day MA	RSI of VIX <= 20	4	180	52.78	-0.02%	0.92
Above 200-day MA	RSI of VIX > 20 and <= 40	4	1128	56.21	0.04%	1.19
Above 200-day MA	RSI of VIX > 40 and <= 60	4	1075	54.79	0.02%	1.07
Above 200-day MA	RSI of VIX > 60 and <= 80	4	607	56.18	0.06%	1.18
Above 200-day MA	RSI of VIX > 80 and <= 100	4	212	55.19	0.08%	1.22
Below 200-day MA	RSI of VIX <= 20	4	73	46.58	-0.19%	0.68
Below 200-day MA	RSI of VIX > 20 and <= 40	4	315	55.24	0.10%	1.18
Below 200-day MA	RSI of VIX > 40 and <= 60	4	360	51.39	-0.04%	0.95
Below 200-day MA	RSI of VIX > 60 and <= 80	4	255	52.94	0.06%	1.08
Below 200-day MA	RSI of VIX > 80 and <= 100	4	73	54.79	0.65%	1.98

No surprise here. An overbought VIX suggests an upside edge, but not as strongly as an oversold SPX. Interestingly, the number of instances where VIX registered an RSI(4) reading greater than 80 was the same number of instances that SPX registered an RSI(4) less than 20. It was 285 instances. RSI of SPX showed a higher % Winners (60% vs 55%), larger average move (0.31% vs 0.23%), and a better profit factor (1.63 vs 1.51). Using RSI to measure overbought/oversold VIX, could identify opportune times to take short-term trades to the long side. But simply using the same measures on SPX appears to provide a more reliable edge.

Of course RSI is just one measure of overbought/oversold. Next let's look at an even simpler one. Here we are going to examine whether SPX (and later VIX) closed at a multi-day high, a multi-day low, or somewhere in between. I'll also delineate "in between" into two buckets using a moving average. So the 4 quadrants are 1) at an X-day low, 2) below the X-day moving average (but above an X-day low), 3) above the X-day moving average (but below an X-day high), and 4) at an X-day high.

The table below uses 10 for X.

Next Day SPX Performance Filtered by High-Low-Upper-Lower Quandrant 10-Day Close and 200ma Filter. 2007 - 2023.

Long-Term Status	Short-term Setup	Xdays	# Trades	% of Winners	Avg % Profit/Loss	Profit Factor
0 Long-Term Filter	1 - Lowest Close	10	554	58.3	0.2	1.4
0 Long-Term Filter	2 - Below Avg Close	10	1078	54.55	0.02	1.05
0 Long-Term Filter	3 - Above Avg Close	10	1532	54.83	0.04	1.11
0 Long-Term Filter	4 - Highest Close	10	1114	50.72	-0.04	0.86

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After seeing the RSI results, these results are no surprise. Oversold suggests an upside edge, and overbought a mild downside edge. Breaking it out by the 200ma also tells a similar story.

Next Day SPX Performance Filtered by High-Low-Upper-Lower Quandrant 10-Day Close and 200ma Filter. 2007 - 2023.

Long-Term Status	Short-term Setup	Xdays	# Trades	% of Winners	Avg % Profit/Loss	Profit Factor
Above 200-day MA	1 - Lowest Close	10	317	60.57	0.12	1.32
Above 200-day MA	2 - Below Avg Close	10	712	55.34	0.03	1.08
Above 200-day MA	3 - Above Avg Close	10	1216	55.02	0.03	1.1
Above 200-day MA	4 - Highest Close	10	957	51.41	0	1.02
Below 200-day MA	1 - Lowest Close	10	237	55.27	0.31	1.45
Below 200-day MA	2 - Below Avg Close	10	366	53.01	0.01	1.02
Below 200-day MA	3 - Above Avg Close	10	316	54.11	0.09	1.14
Below 200-day MA	4 - Highest Close	10	157	46.5	-0.28	0.56

Closing at a low suggests an upside edge in either case. But a low close below the 200ma shows greater potential gains, while a low close above the 200ma shows more reliable gains. Again, this is similar to what we saw using RSI.

Next is a similar breakdown, but using VIX readings as the indicator, rather than SPX readings.

Next Day SPX Performance Filtered by **VIX** High-Low-Upper-Lower Quandrant 10-Day Close. 2007 - 2023.

Long-Term Status	Short-term Setup	Xdays	# Trades	% of Winners	Avg % Profit/Loss	Profit Factor
0 Long-Term Filter	0 Filter - All Days	10	4278	54.14	0.04	1.09
0 Long-Term Filter	1-Highest VIX Close	10	586	55.8	0.18	1.45
0 Long-Term Filter	2-Above Avg VIX Close	10	1298	53.62	-0.02	0.97
0 Long-Term Filter	3-Below Avg VIX Close	10	1549	54.36	0.02	1.07
0 Long-Term Filter	4-Lowest VIX Close	10	845	53.37	0.04	1.14

Here we see a high VIX suggests an upside edge for SPX, while a low VIX does not appear to suggest any edge. Comparing the 10-day low SPX close to the 10-day high VIX close you'll note that the number of instances is similar, but the low-SPX setup shows a higher win rate and a higher Avg % Profit/Loss. Here is the 200ma breakout like we just did with SPX:

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Next Day SPX Performance Filtered by **VIX** High-Low-Upper-Lower Quandrant 10-Day Close and 200ma Filter. 2007 - 2023.

Long-Term Status	Short-term Setup	Xdays	# Trades	% of Winners	Avg % Profit/Loss	Profit Factor
SPX Above 200ma	1-Highest VIX Close	10	425	54.59	0.08	1.24
SPX Above 200ma	2-Above Avg VIX Close	10	940	55	0.02	1.07
SPX Above 200ma	3-Below Avg VIX Close	10	1208	53.97	0.00	0.99
SPX Above 200ma	4-Lowest VIX Close	10	629	55.01	0.07	1.32
SPX Below 200ma	1-Highest VIX Close	10	161	59.01	0.45	1.74
SPX Below 200ma	2-Above Avg VIX Close	10	358	50	-0.12	0.86
SPX Below 200ma	3-Below Avg VIX Close	10	341	55.72	0.11	1.18
SPX Below 200ma	4-Lowest VIX Close	10	216	48.61	-0.03	0.96

Again below the 200ma shows more extreme readings. I will note that a "lowest" VIX reading when SPX is below the 200ma *does* show a negative forward return for SPX, but not nearly as negative as a "highest" SPX reading showed (-0.03% vs -0.28% average loss).

Lest there is concern that the 10-day period was cherry-picked, below is performance for "Lowest" and "Highest" closes for 5, 10, 15, 20, and 25 days. First for SPX, then for VIX.

Next Day SPX Performance Filtered for Times SPX Closed at an X-Day Low or High. 2007 - 2023.

Long-Term Status	Short-term Setup	Xdays	# Trades	% of Winners	Avg % Profit/Loss	Profit Factor
0 Long-Term Filter	1 - Lowest Close	5	929	56.84	0.14	1.31
0 Long-Term Filter	1 - Lowest Close	10	554	58.3	0.2	1.4
0 Long-Term Filter	1 - Lowest Close	15	430	56.98	0.2	1.39
0 Long-Term Filter	1 - Lowest Close	20	356	56.46	0.23	1.42
0 Long-Term Filter	1 - Lowest Close	25	299	56.52	0.27	1.45
0 Long-Term Filter	4 - Highest Close	5	1442	51.66	-0.02	0.95
0 Long-Term Filter	4 - Highest Close	10	1114	50.72	-0.04	0.86
0 Long-Term Filter	4 - Highest Close	15	993	50.35	-0.02	0.9
0 Long-Term Filter	4 - Highest Close	20	904	50.66	-0.02	0.92
0 Long-Term Filter	4 - Highest Close	25	852	51.06	-0.01	0.95

Next Day SPX Performance Filtered for Times VIX Closed at an X-Day Low or High. 2007 - 2023.

Long-Term Status	Short-term Setup	Xdays	# Trades	% of Winners	Avg % Profit/Loss	Profit Factor
0 Long-Term Filter	1-Highest VIX Close	5	978	55.62	0.15	1.39
0 Long-Term Filter	1-Highest VIX Close	10	586	55.8	0.18	1.45
0 Long-Term Filter	1-Highest VIX Close	15	445	55.06	0.16	1.35
0 Long-Term Filter	1-Highest VIX Close	20	348	55.75	0.22	1.46
0 Long-Term Filter	1-Highest VIX Close	25	293	54.61	0.22	1.42
0 Long-Term Filter	4-Lowest VIX Close	5	1289	53.84	0.04	1.11
0 Long-Term Filter	4-Lowest VIX Close	10	845	53.37	0.04	1.14
0 Long-Term Filter	4-Lowest VIX Close	15	668	52.54	0.03	1.11
0 Long-Term Filter	4-Lowest VIX Close	20	560	51.61	0.02	1.08
0 Long-Term Filter	4-Lowest VIX Close	25	505	51.88	0.01	1.04

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The lessons are the same no matter the timeframe. 1) Oversold SPX and overbought VIX both suggest an upside edge. The edge provided by SPX appears to be stronger. 2) Overbought SPX suggests a downside edge, while oversold VIX is more neutral.

You may find it curious that the downside edge for overbought SPX does not appear to increase when looking at more extreme readings (a 25-day high vs a 5-day high). This is because the 20 and 25-day high readings are so much less common below the 200ma where the edge primarily exists. You can see this in the table below that contains the 200ma breakdown.

Next Day SPX Performance Filtered for Times SPX Closed at an X-Day Low or High.
200ma Filter also Applied. 2007 - 2023.

Long-Term Status	Short-term Setup	Xdays	# Trades	% of Winners	Avg % Profit/Loss	Profit Factor
Above 200-day MA	1 - Lowest Close	5	592	57.26	0.06	1.17
Above 200-day MA	1 - Lowest Close	10	317	60.57	0.12	1.32
Above 200-day MA	1 - Lowest Close	15	223	59.64	0.12	1.3
Above 200-day MA	1 - Lowest Close	20	167	59.28	0.11	1.3
Above 200-day MA	1 - Lowest Close	25	121	58.68	0.12	1.29
Above 200-day MA	4 - Highest Close	5	1177	52	0	1.02
Above 200-day MA	4 - Highest Close	10	957	51.41	0	1.02
Above 200-day MA	4 - Highest Close	15	872	51.26	0.01	1.07
Above 200-day MA	4 - Highest Close	20	809	51.42	0.02	1.09
Above 200-day MA	4 - Highest Close	25	775	51.87	0.02	1.11
Below 200-day MA	1 - Lowest Close	5	337	56.08	0.29	1.45
Below 200-day MA	1 - Lowest Close	10	237	55.27	0.31	1.45
Below 200-day MA	1 - Lowest Close	15	207	54.11	0.3	1.44
Below 200-day MA	1 - Lowest Close	20	189	53.97	0.33	1.47
Below 200-day MA	1 - Lowest Close	25	178	55.06	0.37	1.5
Below 200-day MA	4 - Highest Close	5	265	50.19	-0.11	0.83
Below 200-day MA	4 - Highest Close	10	157	46.5	-0.28	0.56
Below 200-day MA	4 - Highest Close	15	121	43.8	-0.3	0.51
Below 200-day MA	4 - Highest Close	20	95	44.21	-0.31	0.51
Below 200-day MA	4 - Highest Close	25	77	42.86	-0.32	0.46

Note the downside edge continues to increase as you make higher-level highs below the 200ma (bottom 5 rows). But the instances shrink enough that the total impact is dulled when lumped together with instances above the 200ma.

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Whether we use short-term RSI readings or whether we simply look at short to intermediate-term highs and lows to gauge overbought/oversold, the message appears to be the same.

- Oversold SPX readings suggest a next-day upside edge.
- The edge is more reliable above the 200ma, but more powerful below it.
- An overbought VIX provides a similar edge, but the overbought VIX stats are not as impressive.
- Overbought SPX readings suggest a downside edge, particularly when SPX is in a long-term downtrend (below its 200ma).
- Oversold VIX readings don't show much of a downside edge. If it appears at all, it is much less pronounced than the downward edge an overbought SPX suggests.

In general, the VIX could be used to design short-term models to trade the SPX. But it is not clear that using VIX readings instead of SPX readings would enhance model returns. In fact, the more robust edges seem to simply be based on SPX movement itself.

Considering VIX-based Securities for Trading

As discussed earlier, VIX readings appear highly correlated to, and perhaps dependent on SPX action. High VIX readings are unlikely without high realized volatility in SPX. And low VIX readings are extremely unlikely when SPX is highly volatile. So rather than use VIX movement to identify edges for trading SPX, would there be more value in the converse? Could SPX movement be used to trade VIX-based securities?

First, let's recall that the VIX itself is not tradeable. But there are VIX futures, options, and exchange-traded products available for trading. So let's examine how some of these products have traded over time.

Below is a split-adjusted look at UVXY, which is the VIX-based ETF that has had the longest continuous history.

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With the value dropping from over $24 billion in 2011 down to $8.44 at the end of 2023 there is a clear long-term downside edge. Of course, some very sharp countertrend rallies happened during big market selloffs. Still, a move from $24 million down to $8 seems like one that could be taken advantage of. And what is especially encouraging is that there was never a long, extended period between new lows.

Notable about UVXY is that the leverage changed from 2x to 1.5x on Feb 28, 2018, after "Volmageddon" in early February wreaked havoc on several VIX-based ETFs. UVXY went from a low of 4260 to a high of 15,090 (354% gain) during that period. During March of 2020 UVXY rose as much as 1300%. I am not going to get into position sizing and risk management in this paper, but it is clearly something that investment managers need to take into account when trading VIX-based products. Still, the long-term edge appears sizable.

Trading the ETFs and ETNs also includes counter-party risk, borrow costs, available shares for shorting, margin requirements, and ETF structure issues. These all create complications that I am not inclined to delve into in this paper, but have discussed elsewhere.

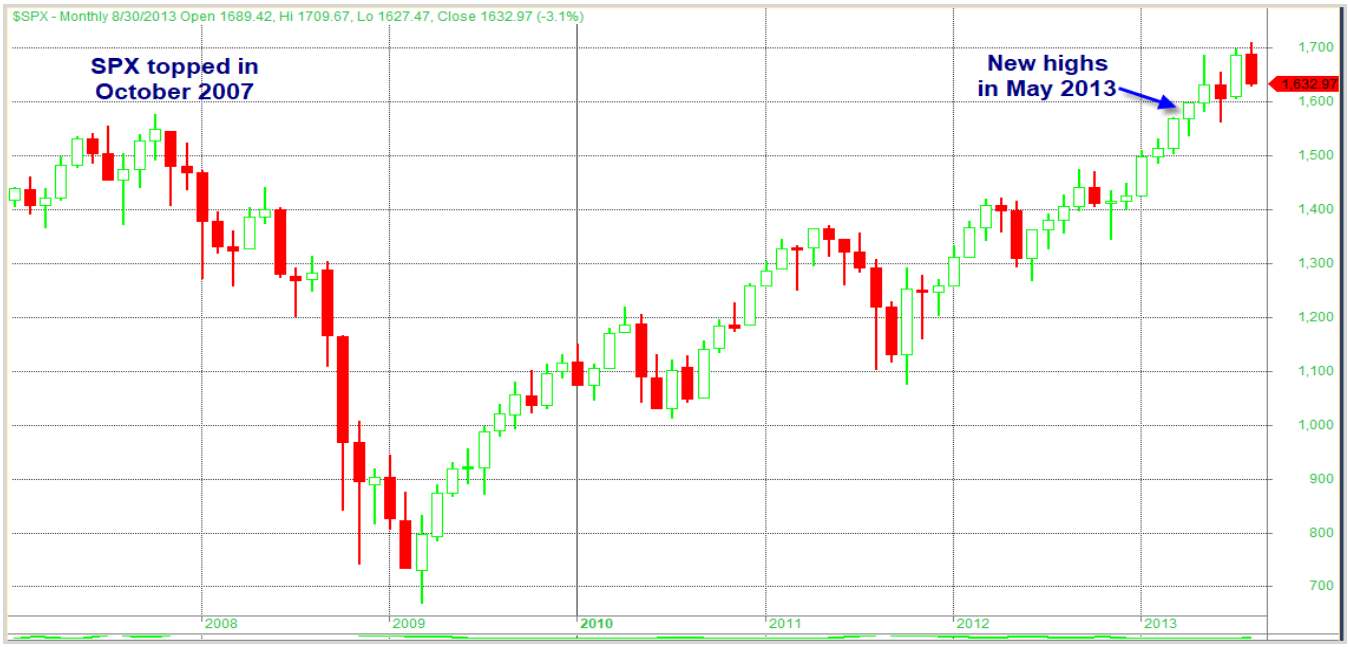
Most of these issues do not exist when looking at VIX futures (VX). Additionally, VIX futures have a longer history. And as shown below, it is not just the ETFs and ETNs, but also the futures, that have shown a long-term downside tendency.

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This continuous futures contract was created with data available directly from the CBOE website. VIX futures expire on Wednesday mornings, so the roll from one month to the next was done at the close on Tuesday just before the Wednesday expiration. The price of the future declined from a roll-adjusted 229.9 on October 31, 2006 to 13.05 at the end of 2023. While there were sharp counter-trend moves along the way, the persistency was impressive. The largest and longest countertrend move occurred during 2008. Next we will zoom in on the 2007-2008 bear market as well as the 2022 bear market.

This chart shows the S&P 500 during the 2007 to 2009 bear market.



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As you can see the S&P topped in October of 2007. The bear market bottomed in 2009, but it wasn't until May 2013 that SPX managed to make a new high. So a buy-and-hold S&P 500 strategy would have spent about 5 1/2 years in a drawdown during this bear market. But let's look and see what happened to VIX futures during this same period.



We see here that the bottom for the VIX futures did not come until May of 2008, a full 7 months after the initial S&P market top. There was a big spike during the bear market, but the recovery was much quicker. VIX futures made a new low in February of 2010. So rather than a 5 ½ year drawdown, a VX "short and hold" strategy would have only endured about a 1 ½ year drawdown. That's a massive difference, and it makes the idea of trading VIX-based products very appealing. When market shocks occur, you don't need a large SPX rally for a short-volatility strategy to make a new high. You just need the market to calm down enough to allow the downside tendency of VIX futures to reassert itself. Next let's look at the 2022 bear market.

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The top panel is the S&P 500 from the beginning of 2022 through 2023. While SPX bottomed in October of 2022, it took until early 2024 to get back to a new high. It was a 2-year drawdown. The bottom panel shows VIX futures. VX barely went six months without making a new low. By the end of July 2022 it was already at new lows, and SPX had not even bottomed yet! When the SPX finally bottomed in October 2022, VX was just weeks away from another round of new lows. This persistent downside tendency creates a massive edge, and the shortened drawdowns make the thought of trading VIX-based products quite appealing vs trading the SPX.

Filtering VX Movement Using SPX/VIX Action

I established earlier that SPX is a better indicator than VIX for anticipating SPX movement. But what about VX movement? Next is a series of studies that utilize SPX readings and VIX readings and measure VX movement based on the same filters we used in evaluating SPX edges. It is important to keep in mind that since the long-term VX trend is down, these studies look at results of going SHORT VX. So a positive result means that VX has exhibited a downside edge under these circumstances. A negative result would indicate times the VX has risen.

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Daily return stats shorting 1 @VX future using RSI(2) of SPX as a filter.
1/1/2007 - 12/31/2023.

Long-Term Status	Short-term Setup	rsiLength	# Trades	% of Winners	Avg Profit/Loss	Net Profit	Max. Sys Drawdown	Recovery Factor	Profit Factor
0 Long-Term Filter	0 ST Filter	2	4278	58.11	0.05	213.52	-68.86	3.1	1.13
0 Long-Term Filter	RSI <= 20	2	822	62.9	0.22	178.6	-21.04	8.49	1.47
0 Long-Term Filter	RSI > 20 and <= 40	2	635	59.37	0.07	44.46	-36.87	1.21	1.16
0 Long-Term Filter	RSI > 40 and <= 60	2	603	59.87	-0.03	-18.18	-45.3	-0.4	0.94
0 Long-Term Filter	RSI > 60 and <= 80	2	851	59.69	0.06	54.21	-25.23	2.15	1.2
0 Long-Term Filter	RSI > 80 and <= 100	2	1367	52.89	-0.03	-45.57	-59.13	-0.77	0.89

These results are intriguing. We see that short-term oversold readings show favorable results and short-term overbought readings are unfavorable for the "short VX" trade. Without any filters, for the full period measured, VX declined 213.52 points (top row). By simply shorting VX when SPX posted an RSI(2) <= 40, and sitting out the rest of the time, more than all of the 213.52 downside points would have been realized. And that would have been accomplished by only being in the market 34% of the time! Also impressive is the fact that RSIs over 80 (bottom row) showed a 45.57 point *gain* in VX over time. This suggests that periods when SPX is short-term overbought are dangerous times to short VX, and they could even provide opportunities for taking long VX positions.

What if we use the RSI of VIX instead of SPX?

Daily return stats shorting 1 @VX future using RSI(2) of VIX as a filter.
1/1/2007 - 12/31/2023.

Long-Term Status	Short-term Setup	rsiLength	# Trades	% of Winners	Avg Profit/Loss	Net Profit	Max. Sys Drawdown	Recovery Factor	Profit Factor
0 Long-Term Filter	RSI of VIX > 80 and <= 100	2	830	58.19	0.11	90.81	-33.12	2.74	1.24
0 Long-Term Filter	RSI of VIX > 60 and <= 80	2	686	57.14	0.05	31.98	-27.48	1.16	1.11
0 Long-Term Filter	RSI of VIX > 40 and <= 60	2	740	57.3	-0.03	-19.04	-58.22	-0.33	0.94
0 Long-Term Filter	RSI of VIX > 20 and <= 40	2	914	58.75	0.03	29.54	-36.87	0.8	1.09
0 Long-Term Filter	RSI of VIX <= 20	2	1108	58.66	0.07	80.23	-24.22	3.31	1.25
0 Long-Term Filter	0 ST Filter	2	4278	58.11	0.05	213.52	-68.86	3.1	1.13

As with the SPX tests we ran earlier, the edges here are not as consistent, nor as strong. An overbought VIX reading (> 80) suggests a favorable condition for "short VX". But the Avg Profit/Loss and Net Profits are about half of what is seen in the SPX RSI <= 20 results from the previous table. Also notable is that while an overbought SPX reading suggests a long VX edge, an oversold VIX reading does NOT. Using RSI of VIX as a filter does not appear nearly as valuable as using RSI of SPX.

I also broke out the returns by whether SPX closed above or below its 200ma. Here is the RSI of SPX breakdown.

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Daily return stats shorting 1 @VX future using 1) whether SPX is above or below its 200ma, and 2) RSI(2) of SPX as filters.
1/1/2007 - 12/31/2023.

Long-Term Status	Short-term Setup	rsiLength	# Trades	% of Winners	Avg Profit/Loss	Net Profit	Max. Sys Drawdown	Recovery Factor	Profit Factor
Above 200-day MA	0 ST Filter	2	3202	57.9	0.04	130.75	-29.84	4.38	1.14
Above 200-day MA	RSI <= 20	2	500	64.6	0.14	68.94	-17.89	3.85	1.35
Above 200-day MA	RSI > 20 and <= 40	2	453	57.84	0.08	38.29	-15.66	2.45	1.25
Above 200-day MA	RSI > 40 and <= 60	2	443	59.82	0.01	6.37	-22.41	0.28	1.04
Above 200-day MA	RSI > 60 and <= 80	2	677	59.97	0.05	30.72	-13.35	2.3	1.17
Above 200-day MA	RSI > 80 and <= 100	2	1129	52.97	-0.01	-13.57	-32.94	-0.41	0.95
Below 200-day MA	0 ST Filter	2	1076	58.74	0.08	82.77	-68.86	1.2	1.12
Below 200-day MA	RSI <= 20	2	322	60.25	0.34	109.66	-21.04	5.21	1.61
Below 200-day MA	RSI > 20 and <= 40	2	182	63.19	0.03	6.17	-36.87	0.17	1.05
Below 200-day MA	RSI > 40 and <= 60	2	160	60	-0.15	-24.55	-42.26	-0.58	0.83
Below 200-day MA	RSI > 60 and <= 80	2	174	58.62	0.13	23.49	-23.12	1.02	1.24
Below 200-day MA	RSI > 80 and <= 100	2	238	52.52	-0.13	-32	-44.95	-0.71	0.77

No surprises. Oversold in an uptrend is more consistent than oversold in a downtrend, but the instances below the 200ma show larger average moves.

Also notable is that the "upside VX" edge almost entirely plays out when SPX is in a downtrend (bottom row). Overbought in an uptrend stats only show small losses when shorting VX.

Next is the 200ma filter breakdown using RSI of VIX:

Daily return stats shorting 1 @VX future using 1) whether SPX is above or below its 200ma, and 2) RSI(2) of VIX as filters.
1/1/2007 - 12/31/2023.

Long-Term Status	Short-term Setup	rsiLength	# Trades	% of Winners	Avg Profit/Loss	Net Profit	Max. Sys Drawdown	Recovery Factor	Profit Factor
Above 200-day MA	RSI of VIX > 80 and <= 100	2	610	57.38	0.04	23.17	-34.42	0.67	1.09
Above 200-day MA	RSI of VIX > 60 and <= 80	2	499	57.92	0.07	37.39	-10.82	3.46	1.25
Above 200-day MA	RSI of VIX > 40 and <= 60	2	551	56.62	0.01	2.91	-18.43	0.16	1.02
Above 200-day MA	RSI of VIX > 20 and <= 40	2	708	58.33	0.04	25.25	-15.58	1.62	1.13
Above 200-day MA	RSI of VIX <= 20	2	834	58.75	0.05	42.03	-24.56	1.71	1.22
Above 200-day MA	0 ST Filter	2	3202	57.9	0.04	130.75	-29.84	4.38	1.14
Below 200-day MA	RSI of VIX > 80 and <= 100	2	220	60.45	0.31	67.64	-23.74	2.85	1.5
Below 200-day MA	RSI of VIX > 60 and <= 80	2	187	55.08	-0.03	-5.41	-26.54	-0.2	0.96
Below 200-day MA	RSI of VIX > 40 and <= 60	2	189	59.26	-0.12	-21.95	-49.84	-0.44	0.84
Below 200-day MA	RSI of VIX > 20 and <= 40	2	206	60.19	0.02	4.29	-33.39	0.13	1.03
Below 200-day MA	RSI of VIX <= 20	2	274	58.39	0.14	38.2	-11.76	3.25	1.28
Below 200-day MA	0 ST Filter	2	1076	58.74	0.08	82.77	-68.86	1.2	1.12

The only strong edge here appears to be overbought VIX readings when SPX is in a long-term downtrend (blue outlined row). In looking at high VIX readings when SPX is above the 200ma (top row), the Avg Profit/Loss is the same as without any RSI filter above the 200ma. And the Drawdown and Profit Factor stats are *worse*.

This all suggests that RSI of SPX is a much better indicator of forward VX performance than RSI of VIX.

This last RSI table shows returns using 2, 3, and 4-period RSIs of SPX.

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Daily return stats shorting 1 @VX future using RSI(2), RSI(3), or RSI(4) of SPX as a filter.
1/1/2007 - 12/31/2023.

Long-Term Status	Short-term Setup	rsiLength	# Trades	% of Winners	Avg Profit/Loss	Net Profit	Max. Sys Drawdown	Recovery Factor	Profit Factor
0 Long-Term Filter	0 ST Filter	2	4278	58.11	0.05	213.52	-68.86	3.1	1.13
0 Long-Term Filter	RSI <= 20	2	822	62.9	0.22	178.6	-21.04	8.49	1.47
0 Long-Term Filter	RSI > 20 and <= 40	2	635	59.37	0.07	44.46	-36.87	1.21	1.16
0 Long-Term Filter	RSI > 40 and <= 60	2	603	59.87	-0.03	-18.18	-45.3	-0.4	0.94
0 Long-Term Filter	RSI > 60 and <= 80	2	851	59.69	0.06	54.21	-25.23	2.15	1.2
0 Long-Term Filter	RSI > 80 and <= 100	2	1367	52.89	-0.03	-45.57	-59.13	-0.77	0.89
0 Long-Term Filter	0 ST Filter	3	4278	58.11	0.05	213.52	-68.86	3.1	1.13
0 Long-Term Filter	RSI <= 20	3	457	64.33	0.27	125.44	-19.34	6.49	1.54
0 Long-Term Filter	RSI > 20 and <= 40	3	782	59.08	0.08	62.12	-42.23	1.47	1.17
0 Long-Term Filter	RSI > 40 and <= 60	3	944	59.11	-0.01	-7.42	-46.77	-0.16	0.98
0 Long-Term Filter	RSI > 60 and <= 80	3	1161	59.6	0.07	75.58	-25.52	2.96	1.21
0 Long-Term Filter	RSI > 80 and <= 100	3	934	51.39	-0.05	-42.2	-60.01	-0.7	0.83
0 Long-Term Filter	0 ST Filter	4	4278	58.11	0.05	213.52	-68.86	3.1	1.13
0 Long-Term Filter	RSI <= 20	4	285	64.21	0.3	84.78	-19.34	4.38	1.52
0 Long-Term Filter	RSI > 20 and <= 40	4	820	59.15	0.09	70.75	-39.63	1.79	1.17
0 Long-Term Filter	RSI > 40 and <= 60	4	1145	59.48	0.04	43.94	-51.57	0.85	1.09
0 Long-Term Filter	RSI > 60 and <= 80	4	1405	59.29	0.04	60.25	-25.6	2.35	1.15
0 Long-Term Filter	RSI > 80 and <= 100	4	623	48.8	-0.07	-46.2	-55.18	-0.84	0.71

The stats change some depending on the RSI length you use, but the implications remain the same. When SPX is short-term oversold, it is a favorable condition for shorting VX. And when SPX is overbought, VX shorting becomes dangerous, and long VX positions could even be considered.

Next let's look at times SPX closed at a high, a low, or in between. This series of tests is set up similar to the series shown earlier that examined SPX performance based on where SPX/VIX closed in relation to a new high, low, or in between. The first table below looks at SPX finishing hi-low-mid using a 10-day measure.

Daily return stats shorting 1 @VX future if SPX is 1) at Highest 10-day close, 2) Above 10-day MA but < Highest,
3) Below 10-Day MA but above Lowest, or 4) at Lowest 10-day close . 1/1/2007 - 12/31/2023.

Long-Term Status	Short-term Setup	Xdays	# Trades	% of Winners	Avg Profit/Loss	Net Profit	Max. Sys Drawdown	Recovery Factor	Profit Factor
0 Long-Term Filter	0 Filter - All Days	10	4278	58.11	0.05	213.52	-68.86	3.1	1.13
0 Long-Term Filter	1-Highest SPX Close	10	1114	52.6	-0.04	-41.78	-66.83	-0.63	0.86
0 Long-Term Filter	2-Above Avg SPX Close	10	1532	60.31	0.07	104.79	-20.4	5.14	1.22
0 Long-Term Filter	3-Below Avg SPX Close	10	1078	58.63	0.03	33.54	-59.41	0.56	1.06
0 Long-Term Filter	4-Lowest SPX Close	10	554	62.09	0.21	116.97	-22.06	5.3	1.41

Consistent with our other tests, a low SPX suggests a favorable time for shorting VX. A high SPX suggests an unfavorable time for shorting VX. Readings in between "highest" and "lowest" are not far from average returns. With middle quadrants not showing much edge (anywhere), the next several tests will focus just on "highest" and "lowest" readings.

The two tables below look at "short VX" performance broken down by "highest" and "lowest" close in the last "X Days". The first table uses VIX highs and lows, and the 2nd table uses SPX highs and lows.

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Daily return stats shorting 1 @VX future filtered for instances where VIX closes at the lowest X day close or the highest X-day close.
1/1/2007 - 12/31/2023.

Long-Term Status	Short-term Setup	Xdays	# Trades	% of Winners	Avg Profit/Loss	Net Profit	Max. Sys Drawdown	Recovery Factor	Profit Factor
0 Long-Term Filter	1-Lowest VIX Close	5	1289	58.57	0.07	88.09	-24.9	3.54	1.23
0 Long-Term Filter	1-Lowest VIX Close	10	845	57.63	0.07	57.15	-11.15	5.13	1.25
0 Long-Term Filter	1-Lowest VIX Close	15	668	57.49	0.06	42.09	-15.3	2.75	1.25
0 Long-Term Filter	1-Lowest VIX Close	20	560	56.79	0.05	30.45	-12.96	2.35	1.22
0 Long-Term Filter	1-Lowest VIX Close	25	505	56.44	0.05	26.28	-11.25	2.34	1.22
0 Long-Term Filter	4-Highest VIX Close	5	978	58.9	0.14	136.01	-25.01	5.44	1.33
0 Long-Term Filter	4-Highest VIX Close	10	586	59.9	0.18	106.27	-26.54	4	1.4
0 Long-Term Filter	4-Highest VIX Close	15	445	58.43	0.16	71.12	-26.89	2.64	1.31
0 Long-Term Filter	4-Highest VIX Close	20	348	59.77	0.18	63.16	-28.58	2.21	1.31
0 Long-Term Filter	4-Highest VIX Close	25	293	59.04	0.15	42.9	-23.57	1.82	1.23

Daily return stats shorting 1 @VX future filtered for instances where SPX closes at the highest X day close or the lowest X-day close.
1/1/2007 - 12/31/2023.

Long-Term Status	Short-term Setup	Xdays	# Trades	% of Winners	Avg Profit/Loss	Net Profit	Max. Sys Drawdown	Recovery Factor	Profit Factor
0 Long-Term Filter	1-Highest SPX Close	5	1442	54.23	-0.02	-28.49	-56.14	-0.51	0.94
0 Long-Term Filter	1-Highest SPX Close	10	1114	52.6	-0.04	-41.78	-66.83	-0.63	0.86
0 Long-Term Filter	1-Highest SPX Close	15	993	52.06	-0.03	-27.08	-50.76	-0.53	0.89
0 Long-Term Filter	1-Highest SPX Close	20	904	52.43	-0.03	-23.4	-43.93	-0.53	0.89
0 Long-Term Filter	1-Highest SPX Close	25	852	52.7	-0.02	-21.21	-39.44	-0.54	0.89
0 Long-Term Filter	4-Lowest SPX Close	5	929	61.36	0.16	144.04	-37.9	3.8	1.32
0 Long-Term Filter	4-Lowest SPX Close	10	554	62.09	0.21	116.97	-22.06	5.3	1.41
0 Long-Term Filter	4-Lowest SPX Close	15	430	60.93	0.19	82.91	-23.63	3.51	1.35
0 Long-Term Filter	4-Lowest SPX Close	20	356	60.39	0.23	81.78	-21.04	3.89	1.41
0 Long-Term Filter	4-Lowest SPX Close	25	299	59.2	0.23	69.77	-21.04	3.32	1.39

The delineation in the bottom table is clearly better. It is all red near the top and all green near the bottom of the table. High SPX closes are a *bad* time to short VX (see the negative Avg Profit/Loss in the top 5 rows of the bottom table). Low SPX closes create opportune times for shorting VX. This can be seen by the strong Avg Profit/Loss numbers and other stats in the bottom 5 rows of the 2nd table.

Delineation in the top table was not nearly as impressive. High VIX readings showed a moderate edge for shorting the VIX. Low VIX readings did not provide a discernable edge. We keep seeing the same theme repeated.

Lastly, I broke down the SPX numbers to show how they looked above vs below the 200ma. This can be found in the table below.

Daily return stats shorting 1 @VX future filtered for instances where SPX closes 1) above/below 200ma and 2) highest/lowest X day close.
1/1/2007 - 12/31/2023.

Long-Term Status	Short-term Setup	Xdays	# Trades	% of Winners	Avg Profit/Loss	Net Profit	Max. Sys Drawdown	Recovery Factor	Profit Factor
SPX Above 200ma	1-Highest SPX Close	5	1177	54.12	-0.01	-12.4	-38.7	-0.32	0.96
SPX Above 200ma	1-Highest SPX Close	10	957	52.56	-0.01	-6.06	-23.53	-0.26	0.97
SPX Above 200ma	1-Highest SPX Close	15	872	52.29	0	1.49	-18.04	0.08	1.01
SPX Above 200ma	1-Highest SPX Close	20	809	52.53	0	-0.08	-17.57	0	1
SPX Above 200ma	1-Highest SPX Close	25	775	52.9	0	-1.81	-18.9	-0.1	0.99
SPX Above 200ma	4-Lowest SPX Close	5	592	61.49	0.09	52.65	-20.23	2.6	1.22
SPX Above 200ma	4-Lowest SPX Close	10	317	64.35	0.15	48.65	-17.57	2.77	1.35
SPX Above 200ma	4-Lowest SPX Close	15	223	63.68	0.15	33.21	-19.34	1.72	1.31
SPX Above 200ma	4-Lowest SPX Close	20	167	62.28	0.19	32.17	-11.46	2.81	1.44
SPX Above 200ma	4-Lowest SPX Close	25	121	61.16	0.2	24.16	-9.41	2.57	1.42
SPX Below 200ma	1-Highest SPX Close	5	265	54.72	-0.06	-16.09	-33.02	-0.49	0.9
SPX Below 200ma	1-Highest SPX Close	10	157	52.87	-0.23	-35.72	-45.98	-0.78	0.6
SPX Below 200ma	1-Highest SPX Close	15	121	50.41	-0.24	-28.57	-39.64	-0.72	0.59
SPX Below 200ma	1-Highest SPX Close	20	95	51.58	-0.25	-23.32	-33.08	-0.7	0.58
SPX Below 200ma	1-Highest SPX Close	25	77	50.65	-0.25	-19.4	-27	-0.72	0.55
SPX Below 200ma	4-Lowest SPX Close	5	337	61.13	0.27	91.39	-24.32	3.76	1.44
SPX Below 200ma	4-Lowest SPX Close	10	237	59.07	0.29	68.32	-21.04	3.25	1.46
SPX Below 200ma	4-Lowest SPX Close	15	207	57.97	0.24	49.7	-21.04	2.36	1.37
SPX Below 200ma	4-Lowest SPX Close	20	189	58.73	0.26	49.61	-21.04	2.36	1.4
SPX Below 200ma	4-Lowest SPX Close	25	178	57.87	0.26	45.61	-21.04	2.17	1.37

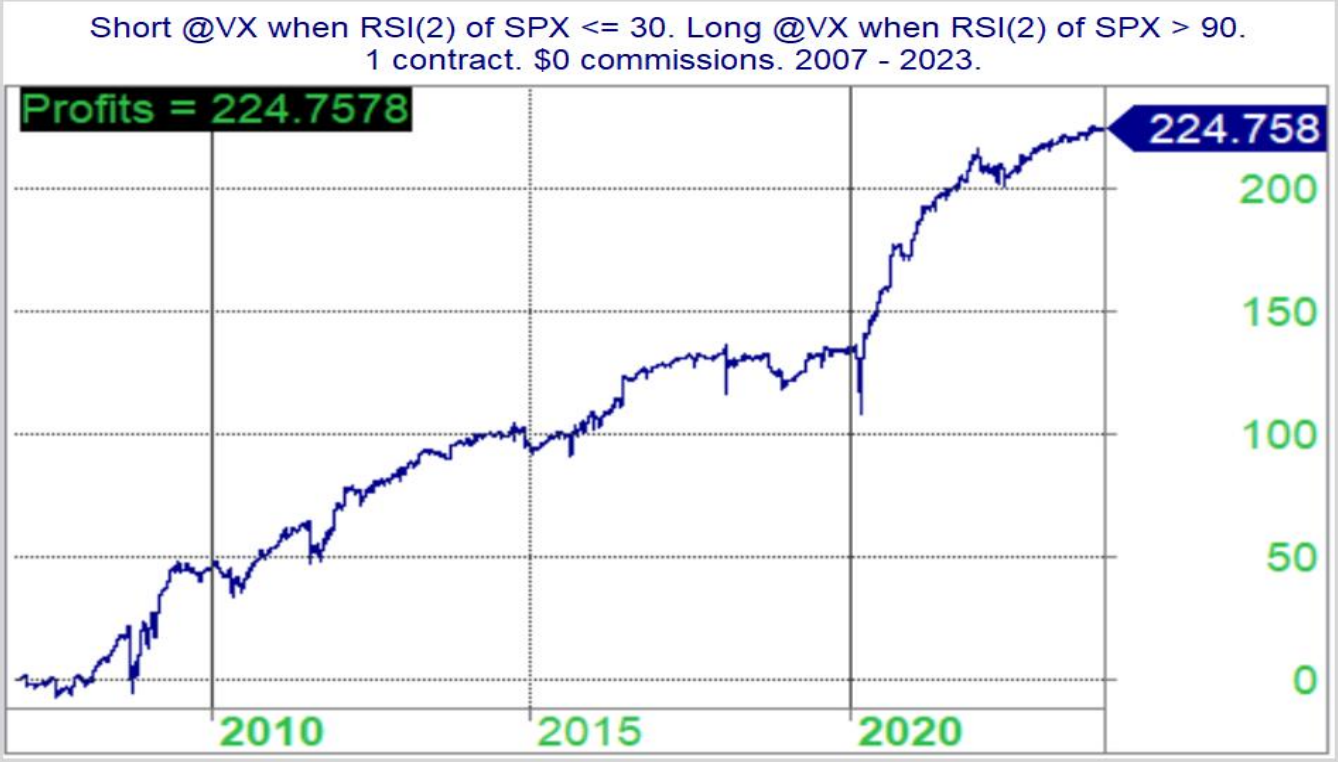
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Like with RSI, 1) strong closes above the 200ma are basically neutral. 2) strong closes below the 200ma show a strong "upside VX" edge, 3) low closes above the 200ma show a high % Winners and muted drawdowns, while 4) low closes below the 200ma show the best Avg Profit/Loss statistics.

A View of a Simple Model

It is clear that there is ample opportunity in shorting VX (and other VIX-based products). The return breakdowns show that SPX action can be used to identify opportune times to take short (and long) VX positions. But while the data tables show net numbers and demonstrate the overall edges quite nicely, the consistency of the edges cannot be seen by simply looking at a data table. For that, a profit curve is helpful. I created a sample model based on the RSI results that were examined earlier. The rules are simple:

1) If the RSI(2) of SPX closes <= 30, short VX on close and hold the next day.
2) If the RSI(2) of SPX closes > 90, buy VX on close and hold the next day.
3) If neither of the above is true, then the position is flat.



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The curve is not perfect, but it is certainly impressive. This was accomplished with just 1 measure (RSI of SPX). There was no filtering for uptrend/downtrend, even though our data suggested that could be used as well. The 225 points of profit are greater than the 213 points that VX lost over the period tested. And these results were accomplished despite only carrying VX exposure 45% of the time. The rest of the time the position would have been flat. Also notable is that the Max Drawdown in the model is 28.06 points. That is much lower than the 68.86 point drawdown for a "short and hold" approach. In fact, it represents a drawdown reduction of 59.25%!

None of this represents a complete system. But it is a solid start, and should help traders and managers consider implications of SPX movement on VIX-based instruments. Of course there are other factors that traders and investment managers will want to consider. These may include, but are not limited to, the VIX futures term structure, upcoming known event risks (elections, economic releases, etc.), and potential seasonality influences, as well as trading costs and tax consequences. Further, position sizing and risk management will need to be taken into account. But it is clear the SPX action can provide a quantifiable edge for trading VX. And it provides even better information than VIX action itself does for trading VX. Utilizing this concept should help traders and investment managers to develop winning strategies.

Conclusion

Lots of energy has been used over the years trying to understand what edges the VIX might provide in predicting SPX movement. While VIX readings can be used to identify compelling setups in many cases, it appears that overbought/oversold measures of the VIX are less predictive than using similar measures of SPX. Additionally, while most traders want to know where the SPX is headed, the easier question is "Where is VX headed?". The downside persistency of VIX-based products make them highly attractive for trading purposes. When a bear market arrives, drawdown lengths are often substantially briefer for a short-VIX position than a long-SPX position. In examining VX behavior, I again find that overbought/oversold measures of SPX provide better filtering than measures of VIX. I believe most traders have been getting it wrong for years. Rather than using the VIX to

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predict SPX movement, more substantial edges lie in using SPX to predict movement of VIX-based products.

According to Morris, Hessels, and Bishops 1968 paper *The relationship between hatching egg weight and subsequent performance of broiler chickens*, "Body weight to 12 weeks was found to be strongly related to egg weight, in a linear fashion, though this influence declined with age." So yeah…bigger eggs get you bigger chickens.

In their paper *Traits influencing the hatching performance of Japanese quail eggs*, Narahari, et al. state "Japanese quail eggs …from moderately heavier dams hatched slightly better than the eggs from lighter dams." They also noted that "Fertility and hatchability were directly proportional to the egg weight." So it appears bigger quails are going to give you a better chance at bigger quail eggs.

If bigger quails make bigger quail eggs, then bigger chickens will most likely make bigger chicken eggs. But thanks to Morris, I am more confident in saying that bigger eggs get you bigger chickens than I am saying bigger chickens get you bigger eggs. Based on my research, I am also more confident in saying that overbought/oversold SPX readings provide a sizable edge in anticipating VIX futures movement than I am saying overbought/oversold VIX readings provide a sizable edge in anticipating SPX movement.

While proper risk management needs to be implemented in any strategy, the techniques shared in this paper can be used to design profitable VX models, and even potentially greatly reduce drawdowns versus "short and hold". When utilizing measures of overbought/oversold on SPX, models can be created that effectively filter unfavorable conditions for attempting short-volatility trades, and can even be used to identify favorable opportunities for taking temporary long VX positions. By turning conventional techniques around, VIX traders can take advantage of the quantifiable edges that SPX movement provides.

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